Experiment 5: TCL Commands — COMMIT, SAVEPOINT, ROLLBACK

AIM

To control transactions using COMMIT, SAVEPOINT, and ROLLBACK commands in MySQL.

Output:

```
mysql> CREATE TABLE Account ( account_id INT PRIMARY KEY, holder_name VARCHAR(50),balance DECIMAL(10,2));
Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO Account
    -> values(1, 'Bhavya', 10000),(2, 'Ravi', 15000);
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Account;
+------------+-------------+----------+
| account_id | holder_name | balance  |
+------------+-------------+----------+
|          1 | Bhavya      | 10000.00 |
|          2 | Ravi        | 15000.00 |
+------------+-------------+----------+
2 rows in set (0.00 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.01 sec)

mysql> UPDATE Account
    -> SET balance = balance - 2000
    -> WHERE account_id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT * FROM Account;
+------------+-------------+----------+
| account_id | holder_name | balance  |
+------------+-------------+----------+
|          1 | Bhavya      |  8000.00 |
|          2 | Ravi        | 15000.00 |
+------------+-------------+----------+
```

```
mysql> SAVEPOINT sp1;
Query OK, 0 rows affected (0.01 sec)

mysql> UPDATE Account
    -> SET balance = balance + 3000
    -> WHERE account_id = 2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM Account;
+------------+-------------+----------+
| account_id | holder_name | balance  |
+------------+-------------+----------+
|          1 | Bhavya      |  8000.00 |
|          2 | Ravi        | 18000.00 |
+------------+-------------+----------+
2 rows in set (0.00 sec)
```

```
mysql> ROLLBACK TO SAVEPOINT sp1;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM Account;
+------------+-------------+----------+
| account_id | holder_name | balance  |
+------------+-------------+----------+
|          1 | Bhavya      |  8000.00 |
|          2 | Ravi        | 15000.00 |
+------------+-------------+----------+
2 rows in set (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM Account;
+------------+-------------+----------+
| account_id | holder_name | balance  |
+------------+-------------+----------+
|          1 | Bhavya      |  8000.00 |
|          2 | Ravi        | 15000.00 |
+------------+-------------+----------+
2 rows in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
```

RESULT

Thus, COMMIT, SAVEPOINT, and ROLLBACK were demonstrated. The update to Bhavya's balance was retained by COMMIT, the update to Ravi's balance was undone by ROLLBACK TO SAVEPOINT, and a separate full ROLLBACK undid an uncommitted update.